

Log 708 exercises

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Log 708 Exercises

This is a collection of exercises for Log 708 - Applied Statistics at Molde University College. Some of the exercises are theoretical questions, while others are practical data exercises where R should be used to solve the problems. The exercises follow the chapter structure of the main text for the course. The level of difficulty ranges from relatively simple to quite difficult, so don't be put down if you can't answer all questions or solve all problems.

Chapter 1

Introduction to statistics

Exercise 1.1. Try to characterize the following variables in terms of the types *numerical and continuous*, *numerical and discrete*, *categorical and ordinal*, *categorical and nominal*.

- The classification of a company in a stock market as belonging to *industry, mining, energy, telecom, transport, etc...*
- The monthly fuel cost for a transport company.
- The main mode of transportation a person used traveling to work on a given day: *car, bus, bike, walk, taxi, subway, passenger in private car, other*.
- The number of transportation modes a person used traveling to work on a given day (For instance, it is 2 if a person went by subway and taxi).
- A stock market analysts recommendation for particular stocks as *Sell, Hold, Buy, Strong buy*.
- The share price of Apple on the New York Stock exchange at a given time.

Exercise 1.2. In a sample of $n = 200$ medical operations at a hospital, the average duration was $\bar{x} = 56$ minutes, with a standard deviation of $S = 11$ minutes. Supposing the distribution of times is fairly symmetrical, find an interval that is likely to include about 95% of the durations of such operations. Discuss how such information can be used for planning purposes at the hospital.

Exercise 1.3. Take a look at the scatterplot in figure 1.1. This shows the urbanization percentage vs the female life expectancy in the sample data `world95`. Consider some country not in the sample.

- Suppose you knew that the female life expectancy was about 50 in the country. What would you guess for the urbanization % in the country? Give an approximate interval where you would expect to find the number.
- Repeat for a country not in the sample, with female life expectancy at 80 years.
- Do the data and the strong correlation indicate that it is more healthy to live in a city than on the countryside?



Figure 1.1: Urbanization and female life expectancy

Chapter 2

Random variables - inferential statistics.

Exercise 2.1. Supposing Z is a standard normal variable, find the probabilities (with three decimals) that

- a. $Z \geq 2$
- b. $Z \leq 3$
- c. $Z \geq 3.93$

Exercise 2.2. Suppose a factory produces some commodity, and to produce one unit of the commodity, they use 3 units of raw material A, and 4 units of material B. If future cost of the materials are uncertain, they can be modeled by random variables X and Y . The *total raw material cost* per unit is then a new variable,

$$W = 3X + 4Y .$$

Assume

$$\mu_x = 200, \mu_y = 90, \sigma_x = 20, \sigma_y = 10 .$$

Also assume X and Y have normal distributions.

Consider the following questions 1 and 2:

1. What is the mean and standard deviation of the total raw material cost?
2. What is the 5% worst-case scenario value for the cost? (I.e. determine a value b such that the cost W is above b with probability 0.05.)

Answer question 1 and 2 in each case a - c below.

- a. The correlation between X and Y is 0.90
- b. X and Y are uncorrelated
- c. The correlation is -0.40 .

Summarize what you find regarding the *risk of high raw material cost* as the correlation shifts from strong positive to zero and further to negative. Try to explain in simple terms why this happens.

Exercise 2.3. Suppose X, Y are two variables with expected value 10 and standard deviation 2. We assume the variables are uncorrelated. Find the expected value and standard deviation for the following

- $W_1 = X + Y$
- $W_2 = 2X$
- $W_3 = X - Y$
- $W_4 = 5 + X$
- $W_5 = 5 + X + Y$

Exercise 2.4. As part of an analysis of the estate market (market for houses and flats) a researcher is interested in the number of days a flat is on the market before it is sold. Let μ be the current mean number of days, and let p be the proportion of flats that were sold within five business days. In a sample of 400 flats, she finds sample mean $\bar{x} = 17$ and sample standard deviation $S = 5$. She also finds that 100 of the flats in the sample were sold within five days.

- a. Find a 95% confidence interval for μ based on the sample data.
- b. Find a 95% confidence interval for p based on the sample data.
- c. In a) you probably found the interval on the form $\bar{x} \pm ME$ where ME is the margin of error. Explain what would happen to the margin of error if (i) We have a larger sample and 95% confidence level. (ii) We have same sample size, but want a 99% confidence interval instead of 95%.
- d. The researcher observes that a few flats are in the market for more than 100 days without being sold. What alternative (to the sample mean) empirical measure would you use for describing “typical number of days to sale” for a flat. In what way could the sample mean be misleading?

Exercise 2.5. This exercise uses data from a random sample of 1600 used cars in Norway. It is about CO₂ emissions from the cars, which is usually measured in grammes/kilometer (g/km). We assume the sample is representative for the whole population of used cars in Norway. In the sample, the mean CO₂ emission was found to be $\bar{x} = 164$ g/km. The sample standard deviation was $s = 35$. Also it was found that 200 of the 1600 observed cars had emissions above 250 g/km. In the following, let μ denote the population mean emissions of all cars, and let p be the population proportion of cars with emissions above 250 g/km.

- a. Find a 95% confidence interval for μ based on the sample data.
- b. Suppose the government has a goal of getting the mean emission μ from cars below 160 g/km. Is it likely that this goal is already reached? (Argue by the confidence interval for μ . If you do not have the answer in a), explain how you would have used the interval).
- c. Find a 95% confidence interval for p .
- d. In the confidence interval above, you use a margin of error (ME). Decide whether the following statements are true or false.
 - i) If the sample size was increased to 2000, with the same sample proportion observed, the ME would be smaller.
 - ii) If the sample size was 1600, but the confidence level was raised to 99%, the ME would be smaller.

Exercise 2.6. Suppose the weekly sales (in liters) of a brand of milk in a supermarket

can be described by a normal random variable with mean $\mu = 900$ and standard deviation $\sigma = 100$. Suppose the supermarket's inventory of the milk can only be increased once a week by reordering.

- a. Find the probability that less than 1050 liters are sold in a given week.
- b. The store manager wants to find an inventory level b so that a stock-out (means the inventory is empty) will occur only in 1 out of 20 (i.e 5%) of the weeks. Determine the level b .
- c. Assume inventory cost for the milk is NOK 200 pr. week. The supermarket earns NOK 1.10 pr. liter sold. The weekly profit from selling the milk is then the revenues from sold milk minus inventory cost. Find the expected value and standard deviation for the weekly profit.

Exercise 2.7. The transportation company OOPS is (among many other things) collecting pallets of empty bottles from supermarkets to bring back to a mineral water producer. Suppose the total time it takes to load a pallet onto a truck at a supermarket can be modeled by a normal random variable with mean 6 minutes and standard deviation 1.5 minutes.

- a. Give an interval of times $[a, b]$ such that the actual loading time at a supermarket will be in the interval with approximately 95% certainty. Describe briefly the rule or method you use.
- b. Find the probability that the loading time is less than 7.5 minutes.



Figure 2.1: Another succesful delivery from OOPS!

Chapter 3

Introduction to R

Exercise 3.1. The first task is to install and get familiar with R and Rstudio. **NOTE: IF YOU ARE WORKING ON A CAMPUS PC - DO NOT TRY TO INSTALL R/Rstudio THERE.** Those PC's already have these programs installed. I.e. the installation part below is only for a PC which does not already run R/Rstudio.

- a. **Installation.** Go to the canvas room for Log 708 and find the video “R_Studio_installation”. This is found in the Panopto Video section, in the subfolder “Videos on basic R”. Note that the instructions in the video are for a Windows PC. If you run on some other platform (e.g. Mac, linux), there may be some minor differences. Check the web for installation instructions before proceeding to install. After completed installation, start Rstudio and check that it looks as in the video. If questions about update appears, choose “ignore update”.

If for some reason, you can not make the R/Rstudio installation work, consider starting with a cloud-based solution here: <https://posit.cloud/>. Here you need to create an account, and then you can run R/Rstudio via a web browser. You get a fair amount of runtime for free, a small fee buys more time. The video “Alternative ways to run Rstudio” shows how this works, along with some other options.

- b. **Folder structure.** Watch the video *More about Rstudio*. Start Rstudio and have a look along with the video to get familiar with the basic interface. Run a few calculations in the console, try to make an R script, a .R file. You should make sure to have a good structure for **all** material relating to your master studies, with a folder for Log 708 in a logical place. Try to establish such a structure now, if you did not already do it. If you don't have a clear picture of how such a structure could look, there is an example folder called **MASTER_Coursework.zip** (alternatively **MASTER_Coursework.7z**) in the **Other Stuff** module in Canvas. You can download and unzip this as a start. Then change and move it as you like. Look to the next point in connection with where you want to have your primary data storage. Note that the **MASTER_coursework** folder is set up with a few extra folders for other courses in the first semester.
- c. **Data safety** In relation to the previous task, think about data safety. In this case we mean “how to avoid losing your work/data”. The college offers a safe disk location where you should store *any material that you really don't want to lose*. Some may have

employers that offers a similar backed-up solution. The local disk on your PC can break down and create a “data disaster”. At least with regular frequency make sure you back up your data in a safe place. So, *the task here* is to figure out in general *how you can store data* relating to your master studies so that you *do not lose them* if your PC breaks down, is stolen or simply refuses to cooperate more with you. The folder structure you made in item b. (or before) should be in such a safe place.

- d. **Make a project for chapter 3 exercises.** Watch the video *Rstudio projects*. It is recommended (though not absolutely necessary) that you collect related work, data and files in an *R project* as described in the video. So now you can make a project with the aim of collecting all your work on exercises for chapter 3. Make sure you have a dedicated folder for work on these exercises. In the setup from part b. there is already such a folder, with path “.../MASTER_coursework/FirstSemester/LOG708/exercises/chapter3”. Here “...” means something that is specific for *your* setup. Now in Rstudio, we choose menu “New project...”. Choose to create project in *existing directory* (“directory” means basically the same as “folder”), and browse to where you want it (e.g. the path above if you use that structure). Then choose “create project” and Rstudio will “switch” to work on your new project. Until you *close* the project, everything you do will be associated with this project. When later you *open* the project again, all your work will be there ready as you left it. In the file menu you find a few commands relevant to project work. You can “open” a project, “open in new session” if you like to keep your current project active while opening another, and you get a list of recently active projects. When you pause work on a project for some time, it is a good habit to *close* the project. In the top right corner of Rstudios window, there is also a list of project, so you can quickly switch, open, or close projects. In the project folder, there will now be a file named `chapter3.Rproj`. This is your project file. You can also activate the project by double-clicking this.
- e. **Setting default working directory** The *default working directory* is a folder that Rstudio/R will generally use as its working directory after startup. If you open a project, the working directory will be changed to the project folder initially. You can now decide what will be your default working directory. If you used the proposed folder structure from Canvas, there is a ready made folder called `myRwork` in the LOG708 folder. This could be a good choice as default working directory. Note that you may change this whenever you want. Here is how you do it in the easiest way to begin with. In Rstudio, go to menu “Tools/Global Options...” In the topic “R general” find the field labelled “Default Working Directory (when not in a project)”. Choose “Browse...” and find/select the folder that you want to use as default working directory. The result will show the *file path* to the working directory in the field text. When you are happy, choose “Apply” and “OK”. The file path is a text (in Windows) of type “M:/folderA/folderB/folderC”, it will look similar on MAC. This particular file path means that we have a drive called “M” and on this drive we refer to “folderC” which is in “folderB” which is in “folderA”. More about this later.

Exercise 3.2 (Variables and vectors). In this exercise we continue with the introductory videos that you find in **Canvas-> Panopto Video -> Videos on basic R**. Also, in the Canvas module “**R intro-video related material**” you find the R scripts that was used along with the videos. You can download a .zip folder with all scripts or one by one.

As you watch videos as suggested below, you could keep the related Rscript open in Rstudio and try out things as you watch. (Pause video). You are of course free to add and change the code in the scripts you downloaded. (The originals will be unchanged in Canvas anyway.)

- a. Watch the video *Variables and Vectors*. As you watch (or after, if you like) make sure that the codes and tricks used in the videos works in the same way with your Rstudio installation. Note, it is not the point to *memorize* everything in these videos, its more about getting started, and then as you gain experience and time with Rstudio, you will start getting hold of the most important things. Using Rscripts ensures you always have a record of how you solved a given task, and you can check back when you forget.
- b. The following shortcuts in Rstudio (for Windows) are very handy. MAC users might need to research for similar alternatives, CMD is often equivalent to CTRL.

```
#keyboard shortcut CTRL+ENTER    (Used with the script editor, executes selected commands)
#keyboard shortcut CTRL+C         (Copy selected text/code)
#keyboard shortcut CTRL+V         (Paste copied text/code)
#keyboard shortcut CTRL+Z         (works in the editor only, will sequentially undo last ty
#Arrow up / down                  (works differently in editor and console)
# ALT+"-"                         (Insert assignment arrow)
```

Start getting used to using these shortcuts now!

- c. In an R script file (with a name relating to which exercise you now work on), copy in the following comments, and write code that executes what is asked for in the command. So for example if the comment asks

```
#assign the number 10 to variable named a, and print the value to the console.
```

you should answer with the following code, which gives the output below in the console.

```
a <- 10
a

## [1] 10
```

Run each answer to check that the code does what it is supposed to do. (recall CTRL + ENTER executes the line where the cursor is in a script file)

Here are some tasks. Copy *all* the comments here into a new script file, then write code below the corresponding comment. You can most easily copy the code by clicking in the top right corner of the shaded text below.

```
# assign your year of birth to a variable "year"

# make a vector "date" = (year, month, day) with your date of birth.
# Recall, there is a super-central R function that creates vectors.

# make a variable "a" that contains the value (2 + 3)*(10 - 3)^2.

#assign the value 10 to variable "b" and let z be the sum of a and b.
```

```

#assign the value 1 to "a", 2 to "b", 3 to "c".
#you can just overwrite the existing content in the variables.

#test (using logical operator) whether a is equal to b.

#test whether a+b = c.

#assign the numbers 12,13,14 and 15 to "f", i.e. make a vector.

#make a vector y that contains the product by b of all elements in f.

#Let u, v be the vectors below:
u <- c(55, 29, 51, 35, 33, 42)
v <- c(2, 4, 6, 8, 10, 12)

#assign the 4th element of u to a variable x

#change the first element of u to the value 25

#make a logical vector z with values TRUE/FALSE depending on whether u < 35

#make a logical vector w with values TRUE/FALSE depending on whether u < 35 and v > 5

#use R functions to find the length and type of u and w

#check that u+v, u*v, u/v and v^4 gives the proper result in vectorized form

#use ls() to list the global environment. Check that the result is the same as you see in the
#with the cursor in the script window, hit CTRL+SHIFT+ENTER. This should run ALL commands #in

#Save your script file with a sensible name in a sensible place, close it.
#Open it again to verify that your work is all there.
#congratulate yourself with the excellent work, and take a short
#break:-)

```

Exercise 3.3 (Functions and sequences). We move on to the video *Functions and Sequences*. Watch this video now. You might want to run Rstudio in parallel, with the accompanying R script open.

Make an R script for the solution of this exercise. As before, copy the tasks in comments below to the script file.

```

#make a vector z of 5-10 arbitrary numbers.

#Find the sum of elements in z

#Find the average value of z

```

```

#Find the median value of z

#find the mean value of z^2

#In the script editor, write "M <- med". What happens? Hit TAB. What happens?

#At the console, write "fact", and use this to find "factorial(10)" which is 10*9*8*...*3*2*1.

#Suppose X is a standard normal variable. Find the probability that X < 2.5

#Find the probability that X > 2.5 (there are two ways, either law of complement or use the l

#Suppose W is a normal variable with mean 4000, standard deviation 300. Find the probability

#Find all probabilities that W < x, for x = 4100, 4200, 4300, 4400, 4500 utilizing vectorizati

#write R code to generate the following sequences

# 1,3,5,7,9,11

# 5,6,7,8,9,10

# 1,3,1,3,1,3,1,3,1,3

# 8,7,6,5,4,3,2,1

#suppose y is the vector defined by
y <- c(5, 6, 12, 53, 15, 95, 12, 51, 25, 34, 59, 15, 40, 34)

#find the length of y

#extract a vector with the last 6 elements of y

```

Exercise 3.4 (Data frames - basics). We continue with the video *Data frames - basics*. After going through the video, do the following tasks in the same way as before.

- We can start with an example data frame similar to the one in the video. Here a few European countries, their population (2024) and GDP in USDB (2023). Data source: Copilot/chatGPT.

```

DF <- data.frame(name = c("Norway", "Sweden", "Denmark", "Finland",
                          "Germany", "Holland", "Switzerland", "Poland"),
                 population = c(5.5, 10.7, 5.9, 5.5, 84.4, 17.6, 8.8, 40),
                 GDP = c(593, 593, 400, 300, 4456, 991, 885, 1801))

#view the dataframe in the console - and in a spreadsheet-like view.

```

```

#use R functions to find number of rows, number of columns.

#get a vector with the names of variables in DF.

#play with head and tail to show first and last 3 lines of DF.

#rename column "GDP" to "gdp"

#Find the median GDP and the mean population in DF.

#Extract the column "population" into a separate vector "pop" (outside of DF)

#Extract subset A of DF with countries of population greater than 8 million

#Extract subset B of DF where population is greater than 8 and gdp < 1000.

#How can you also extract subset B from A?

#extract row 5 from DF

#extract a dataframe with rows 2,3,4 from DF

#extract a dataframe with columns 2, 3 from DF

#extract a dataframe with columns 1 and 3 from DF

#extract a dataframe with rows 1-4 and columns 1 and 3 from DF

#run this code, but try first to guess what it does:
subset(DF, startsWith(name, "S"))

#compute a new column gdppc, containg GDP per capita for the countries.

#given that population is in millions, and GDP is in billions USD, what is correct unit for g

#use "summary" function to summarize the data in DF.

```

- b. The video did not discuss sorting of dataframes, so we look briefly at that topic here. The `order` function is what we use in base R to sort vectors and dataframes. This works as follows: if `x` is a vector, `order(x)` returns the indices of `x` elements from the smallest to the largest. To sort a vector `x` we can do as follows

```

#define x
x <- c(2, 15, 6, 13, 4, 11)
#check order(x)
order(x)

```

```
## [1] 1 5 3 6 4 2
```

```
#write x in increasing order  
x[order(x)]
```

```
## [1] 2 4 6 11 13 15
```

```
#write x in decreasing order  
x[order(x, decreasing = TRUE)]
```

```
## [1] 15 13 11 6 4 2
```

So, to sort a dataframe according to a column `z`, we need to get the order of `z` and use this to reindex the dataframe. So we can sort `DF` by `population` as follows.

```
# get the order of DF$population into S, use S to index DF:  
S <- order(DF$population)  
DF[S, ]
```

```
#the same in just one step  
DF[order(DF$population), ]
```

See if you can use this method to sort `DF` by `gdp`. Then sort by `gdp` in decreasing order. What happens if you sort `DF` by `name`?

Exercise 3.5 (Visualization). This exercise is based on the video *Visualization*, and the related R script `Visualization.R`. We will practice very briefly some of the methods here. We can use the dataframe `DF` defined above to practice. Put your codes and comments in an R script as usual. Use base R as outlined in the video unless you are familiar with `ggplot2`

- Make a scatterplot of `population` vs `gdp`.
- Save the plot as a `.pdf` file.
- The `boxplot` function works pretty much as `plot`. Make a boxplot of the variable `gdppc` that you calculated before. If results are deleted, you should have the code to quickly reproduce the work.
- Try to make a histogram of `gdppc`. The function `hist` makes histograms in base R.
- OPTIONAL: If you want to try out `ggplot` you should install the package, using `install.packages("ggplot2")`. This you need to do only once. Then activate the package using `library(ggplot2)`. This you need to repeat once for each R session. If successful, try to replicate the plots from the script `Visualisation.R`.

Exercise 3.6 (NA's and factors). Now, we can watch the video *NA's and factors*. The relevant R script is `Na_factor.R`. Continue working with an R script.

- Let `DF` be the same dataframe as before (8 countries). Assign the value `NA` to Poland's GDP. (Hint: `DF[i, j]` refers to element at row `i`, column `j`.)
- How can we deal with computations involving the GDP variable when `NA`'s are present. Compute the mean, ignoring `NA`-values.
- Run `summary` on `DF`. What happens when `NA`'s are present?

- d. Suppose the result of 20 tosses of a coin ends up as the vector below. Use R code `table(toss)` to summarize the result, and make a barplot of it. What is missing in these results, and how can you fix it?

```
toss <- c(4,4,3,5,3,6,3,5,4,3,4,3,4,1,3,1,5,5,4,3,4,1,6)
```

- e. What is the probability of getting no 2's in 20 consecutive die tosses?

Exercise 3.7 (Descriptive statistics). In this exercises we (unsurprisingly) focus on material from the video *Descriptive statistics*. You might want to watch that one, as well as opening the related R script `Descriptive_statistics.R`.

In the exercise we will examine one of the built-in datasets in R; named `airquality`. Use the following code to activate the data, and also get a description of the data. The use of a shorthand name `AQ` is optional. (Alternatively the command `data(airquality)` will put the dataframe into the environment, with the full name.)

```
#find information about the data, let the name AQ refer to the dataframe.
?airquality
```

```
AQ <- airquality
```

- Use R code to verify the information about number of observations, number of variables, and name of variables that you got from `?airquality`. Use `head`, `tail`, `View` to get an idea of how the data looks.
- It's a bit ugly to have a variable named `Solar.R`. Try to rename that variable to `SolarRad`.
- Use `summary` to get a key numbers for each variable. Note that there are `NA`'s. How many? In which variables?
- Try to use `tapply` to obtain the mean temperature by month. Make a barplot to visualize.
- Try this:

```
with(AQ, plot(Temp,
              type = "l",
              xlab = "Day",
              main = "Daily temperature (F)",
              lwd = 2))
```

What do we get?

- Suppose we want the grouped mean for Solar Radiation by month. The straight code used in the video script will produce some `NA` results. Try to add `na.rm = TRUE` as a fourth argument to your `tapply` statement, to force calculation of means in presence of `NA`. I.e. something like:

```
with(AQ, tapply(..., ..., ..., na.rm = TRUE))
```

- Store the result from question f in a vector `solradmean` and make a barplot of this. Add

labels and title if you like.

- h. Use the formula $C = \frac{5}{9} \cdot (F - 32)$ to compute the temperatures in degrees Celsius, store this in a new column in AQ.
- i. Make a boxplot of the Celsius temperature for the whole period. Also make a display of boxplots by month. Add labels and title as you like.
- j. Compute the correlation between wind and temperature. Make a scatterplot to visualize. In case you want to compute correlations involving NA's, you need to add `use = "complete.obs"` to your `cor` statement, like:

```
with(AQ, cor(Ozone, Temp, use = "complete.obs"))
```

Try also to get the whole correlation matrix for AQ.

Exercise 3.8 (About working directory, reading data files and internal R help.). Watch the video *Working directory*, and then the video *Data files*, up to about 15 minutes. (The last part deals with excel files and larger csv files and is not relevant now.)

- a. Use the command `getwd()` to find the path of you working directory. This is a string (a text) looking for example like “M:/MASTER_Coursework/FirstSemester/LOG708/exercises/chapter3” if you have used the proposed folder setup from exercise 3.1 a. and put the main folder on a drive called “M”. It will probably look similar - but not exactly like this on you PC.
- b. If you have not already done so, download the compressed folder `log708data.zip` (or `log708data.7z` if you prefer that) from canvas, and extract the files into a logically good place in you LOG 708 folder system. (e.g. into the (empty) `log708data` folder in the suggested structure). Get help (from a person or the web) if you run into problems.
- c. We now want to read from disk the data file `flights_NO.csv`. Follow the procedure in the video using “Import Dataset...” Copy the generated code with `... <- read.csv("...")` into your R script file. You can re-execute the code at any time to read the original data again. Continue writing code in the R script. The file contains data regarding a set on airline flights in Norway. Details are not important here. Use `head(...)` to see the top rows of the data.
- d. Use the code

```
df2 <- subset(..., Day.of.Week == "Monday" & Seats > 50)
```

to make a subset of the whole data set. What does `df2` contain? How many rows are in `df2`, and in the original data? (Here “...” is the name you have for your data read from disk)

- e. You have now successfully read data into R. Let's try to write the data in `df2` to disk: Try

```
#write df2 to disk with given file name.
write.csv(df2, "LargeMondayFlights.csv", row.names = FALSE)

#view the working directory, you should see the new file there:
dir()
```

You can also see your new data file using the operating system (Windows, MacOS).

- f. Read the “LargeMondayFlights.csv” back into a new dataframe `df3`. This can be done with the simple code

```
df3 <- read.csv("LargeMondayFlights.csv")
```

since the file happens to be in your working directory. The way we have done this now leads to `df2` and `df3` having the same data, except for differing row numbers. There are simple ways to ensure that the two dataframes are completely identical, for example we could use this:

```
write.csv(df2, "LargeMondayFlights.csv")
df3 <- read.csv("LargeMondayFlights.csv", row.names=1)
```

Note that the `write.csv` command will overwrite an existing data file without asking. For large dataframes, it can be impossible to manually check that they are identical. There is a function `identical(A, B)` that can be used to compare any two R objects A, B - use this for the dataframes.

- g. See if you can write the data frame `df2` to a file with the name “LargeMondayFlights_NO.csv” in your main data folder. (Hint: copy the relevant part of the path that was used when reading the original data, make sure to change the last part, otherwise you will overwrite the original full data set.)

You now know something more about reading and writing data from/to disk storage. You can delete the newly created data files if you like, but *keep the R code* in your R script. (You are working on an R script for this exercise also, aren’t you ?? :-))

Exercise 3.9 (R Packages). In this exercise we will practice how to install and use R packages.

We will use a built-in data set called `mtcars` along with the `ggplot2` graphics library.

First, to install `ggplot2` run this command:

```
install.packages("ggplot2")
```

You usually need to run this command only once on a given PC. That should *install* the package for later use. You may get some “warnings” from R when doing this, ignore those. If you should happen to get an “error” which you cannot resolve through google or similar, get in touch with someone (teacher, assistant teacher) to try to fix.

Next, in *each* R session where you want to use the package, you need to *activate* it. This is done with the `library` command:

```
library(ggplot2)
```

There may be some warning messages, about R versions and such. Ignore those.

So assuming you now have installed `ggplot2` successfully, let’s get some data. Here we are going to use a data set called `mtcars` just for illustration. Since these data are built-in for R, we can use the `data` function as follows

```
#include "mtcars" as a dataframe in your workspace
```

```
data(mtcars)
```

```
#check top 6 rows
```

```
head(mtcars)
```

```
##           mpg  cyl  disp  hp  drat    wt   qsec  vs  am  gear  carb
## Mazda RX4      21.0   6  160 110  3.90  2.620 16.46  0  1    4    4
## Mazda RX4 Wag  21.0   6  160 110  3.90  2.875 17.02  0  1    4    4
## Datsun 710      22.8   4  108  93  3.85  2.320 18.61  1  1    4    1
## Hornet 4 Drive  21.4   6  258 110  3.08  3.215 19.44  1  0    3    1
## Hornet Sportabout 18.7   8  360 175  3.15  3.440 17.02  0  0    3    2
## Valiant         18.1   6  225 105  2.76  3.460 20.22  1  0    3    1
```

```
#get some info about the data
```

```
?mtcars
```

Without going into detail, we see that the data is about cars and fuel consumption, plus diverse characteristics of the cars. Since the point of this exercise mainly is to see how we can install packages, we will not dig deeper into the data, just check that the `ggplot` function in the package `ggplot2` works as intended. Here is an example that you could try to run. The result should look like the figure below the code.

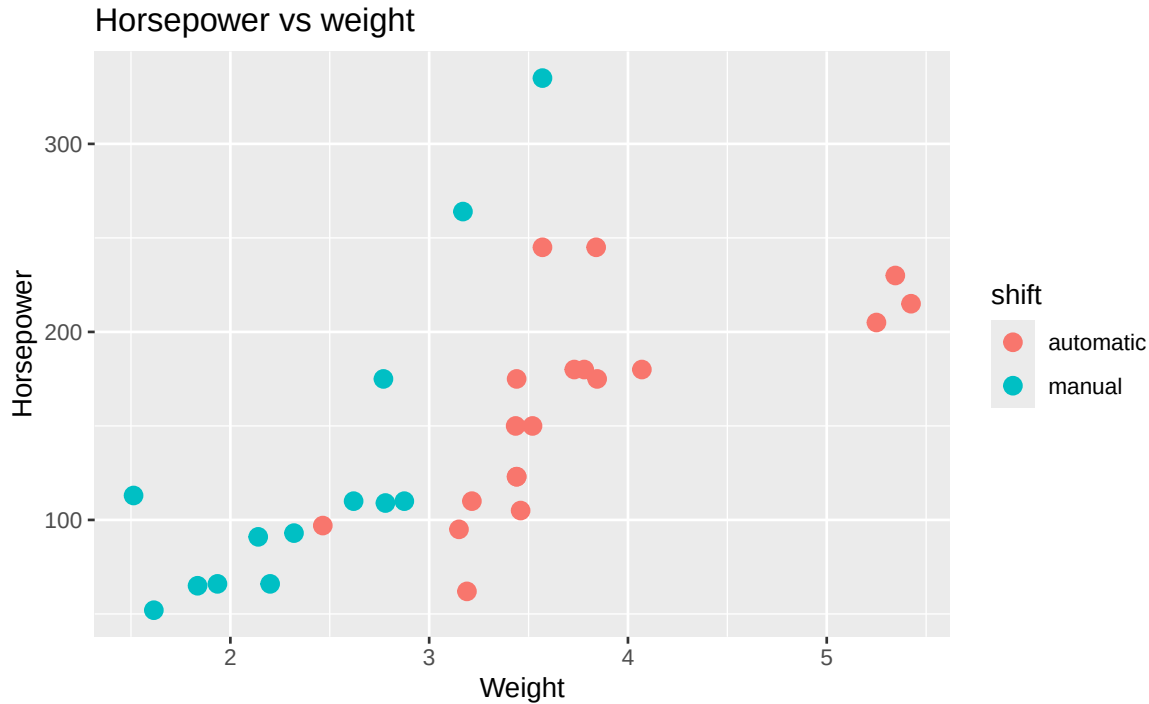
- a. What is roughly the (not unexpected) relationship between weight and horsepower? How is automatic vs manual shift affecting the data? Such visualizations show patterns and relations in data that could otherwise be difficult to spot.

```
#Redefine variable "am" as a factor, just to make figure labels look right.
```

```
shift <- factor(mtcars$am, labels = c("automatic", "manual"))
```

```
#make a plot
```

```
ggplot(mtcars, aes(x = wt, y = hp, color = shift)) +  
  geom_point(size = 3) +  
  labs(title = "Horsepower vs weight",  
        x = "Weight",  
        y = "Horsepower")
```



- b. Add the code `+ geom_smooth(method = lm, se = FALSE)` to the code above. Make sure that the “+” comes on the last line of existing code. Can you guess/describe what the new elements in the figure represents? (Don’t worry if you can’t, we will learn later anyway! :-))

Exercise 3.10 (Getting help). In this little exercise we try out some of R/Rstudio’s internal help tools, as well as external (web) sources. One should note that the internal guides in R are often somewhat compact and technical, while on the web you find more detailed examples.

Suppose we want to learn a bit about how R can help us do calculations with normal distributions. Suppose we have a standard normal variable Z and also another normal variable X with mean = 10, and standard deviation = 3.

We want to find out how we can find such things as

$$P[Z \leq 2.1], P[Z \geq 2.1], P[X \leq 12], P[8.4 \leq X \leq 10.9].$$

- Try to use the internal help system, via `?Normal` or `help(Normal)` to find the probabilities. You should in this case *not* use standardization of X for the latter two cases, since there are built in general normal distributions in R.
- See if you can find information about the same type of calculations via google (or similar). In fact, if you have access to some AI assistant like chatGPT or Copilot, this can be a very good place to ask R-related questions.
- Proceed by internal or external help to find a number b such that $P[Z \leq b] = 0.80$.
- Can you also find a number c such that $P[X \geq c] = 0.90$?

- e. Can you produce a vector `my_x` of 30 random numbers from the distribution of X ? (The same normal distribution). Find the (sample) mean and standard deviation of `my_x`. Are they close to the real parameters?

Chapter 4

Hypothesis testing

In all tests, use significance level $\alpha = 0.05$ unless otherwise specified

Exercise 4.1. In a supermarket, the manager tries to forecast and plan operations for the next quarter. One assumption is that the mean spending μ per customer is 290 NOK. A second assumption is that only 20% of customers pay with cash, while 10% of customers make a cash withdrawal using bank card. In a sample of 400 customers, the average spending was found to be $\bar{x} = 319$ NOK, with a standard deviation of $S_x = 100$. Also in the sample, 56 of the 400 customers payed with cash. We can help the manager test her assumptions - formulated as null hypotheses.

- a. With μ as above, test

$$H_0 : \mu = 290 \text{ vs. } H_1 : \mu \neq 290 ,$$

using 0.05 as significance level. (This is a one-sample t-test, use $N(0, 1)$ as approximate null distribution)

- b. Let p be the proportion of customers paying cash. Test

$$H_0 : p = 0.20 \text{ vs. } H_1 : p \neq 0.20.$$

- c. Suppose there are 10% customers who makes a cash withdrawal, and that their average withdrawal is 400 NOK. Suppose 20% of customers pay with cash. Would you agree that these figures together with the average spending for all customers, imply that the supermarket will generally have a positive cash balance (so as to be able to give back change, and to accommodate withdrawals)?

Exercise 4.2. Find the file `Trip_durations.csv`. These data were used as an example in section 4.5 of the compendium. Read data into a dataframe. Let μ_d be the (true) mean duration of trips, and let μ_s be the mean distance of trips.

- a. Focus on the `Duration` variable. Use R to find the sample size n , mean $\bar{x}$ and standard deviation S_x for the sample in the data file.
- b. Suppose we want to test

$$H_0 : \mu_d = 24 \text{ vs. } H_1 : \mu_d \neq 24$$

based on the data. Write an expression for the test statistic T . What is the null distribution of T ? Compute the observed value T_{OBS} without using the `t.test` function

- in R. Using $N(0, 1)$ as approximate null distribution, find the P-value for the test. Finally, if the significance level is $\alpha = 0.05$, should we reject H_0 ?
- Run the one-sample t-test for the above hypotheses in R. Compare the T_{OBS} value and the P-value you found with R. They should be very similar.
 - Repeat step b) and c) for the hypotheses

$$H_0 : \mu_d = 23 \text{ vs. } H_1 : \mu_d \neq 23 .$$

- Now, focus on the **distance** variable. Does the sample provide significant evidence that $\mu_s > 10$? Use significance level $\alpha = 0.05$. Formulate the appropriate hypotheses and use R to find the P-value.
- Use R to find 95% confidence intervals for the means μ_d, μ_s . Do you see a connection between the conclusions in the tests and the confidence intervals?
- One would expect a very strong correlation between the distance and duration of a trip. Find the correlation coefficient, and make a graphical display of the relation between the variables.

Exercise 4.3. Find the file `flat_prices.csv`. Save a copy. The file has data for sold flats in Molde, Kristiansund and Ålesund. The data is somewhat dated, and does not reflect current market prices. That is not important. We will try some two-sample t-tests to compare prices in the three towns. In all tests, use $\alpha = 0.05$ for the significance level.

- Take some time to familiarize yourself with the data material. The **town** variable is encoded 1 for Molde, 2 for Kristiansund, 3 for Ålesund. Find the mean prices in the three towns. (Hint: in R `tapply(x, g, f)` will apply the function `f` to the variable `x` for each group defined by `g`.)
- Make a boxplot for the prices in the three towns. (Hint: `with(DF, boxplot(x ~ g))` will make a boxplot of `x` for each group in `g`, using data from a dataframe `DF`).
- Compare the mean selling price in Molde and Kristiansund, using a two-sample t test and R. Use the two-sided alternative. You should find that the samples do not provide significant evidence of different mean prices. (Hint: the `t.test` function in R requires the grouping variable to have only two values, but **town** has three. use `subset(DF, town != 3)` to make a new dataframe with only Molde, Kristiansund flats. Use this in the `t.test` as described in the text) Try the same with Kristiansund and Ålesund. In the latter case, you come close to rejecting H_0 but still not all the way.
- Find the sample mean square meter sizes in the three towns. Where are on average the larger flats in the sample? (Hint: same as in a.)
- Study the general correlation between square meter size and selling price in the sample. Make a scatterplot to visualize. (Hint: the function `cor` is useful.)
- Make a new variable, **sqmprice** that measures the *square meter price*, i.e. the selling price, divided by the square meter size. (Hint: `DF$z <- DF$x / DF$y` computes the ratio of `x` to `y` and saves in a column `z` for a dataframe `DF`)
- Now compare the mean square meter prices in Molde and Kristiansund with a two-sample t-test. (Hint: Same as in c, just different variable)
- Try to explain why the results become so different when comparing nominal prices as opposed to comparing square meter prices. (There are at least two reasons involved in this case).

- i. Compare Kristiansund vs. Ålesund, and Molde vs. Ålesund in terms of mean square meter prices.

Exercise 4.4. Work on the data in `flat_prices.csv`. Let's define the parameter p as the (true) proportion of flats sold that have 2 rooms or less. With significance level 0.05, we want to test the hypotheses

$$H_0 : p = 0.50 \text{ vs. } H_1 : p < 0.50 .$$

- a. Try to make a vector `countsm` that shows the number of small flats (less than 3 rooms) and large flats (all others). Hint: make a new variable in your dataframe, `small` indicating whether a flat is small or not, then use `table` to count the occurrences of values for `small`.
- b. Your vector `countsm` from a. should be `[59, 91]` indicating that there are 59 small and 91 not-small flats. Use this with the `binom.test` function to test the proposed hypotheses.

Exercise 4.5. Work on the data in `flat_prices.csv`. We want to check whether the selling prices of houses appear to follow a normal distribution. Follow the outline in the end of chapter 4 of the text when using R for this.

- a. Think of a typical estate market. Would you expect prices to approximately follow a normal distribution? What typical characteristic of such markets could be in conflict with a normal distribution?
- b. Do the normality tests for the price variable in the data set. Comment on your findings. Was your thinking in a) right?
- c. Repeat the testing for normality, now using `town` as a factor. It means you do the test within each town's estate market. How would you explain the difference in results for the towns? (Hint: If your dataframe is `Df` try something like `with(Df, tapply(price, town, qqnorm))` to apply the normal plot function to price for each town. Do something similar with the function `shapiro.test`.
- d. Are the *square meter prices* normally distributed? (Assuming you have computed the new variable as suggested in exercise 4.3).

Chapter 5

Basic Regression Analysis

Exercise 5.1. Consider the basic model assumptions C, D for regression. Try to think of situations with variables X and Y where we may see violations of assumptions C and D, respectively. (Hint: Often the problem comes from using wrong (too simple) models. Think about how a scatterplot might look if C or D fails.) We restate the conditions here:

We consider the linear model formulated by

$$Y_i = \beta_0 + \beta_1 X_i + \mathcal{E}_i ,$$

With this formulation, we can state the basic model assumptions.

C. The error term is statistically independent of X .

D. For all i , the standard deviation is the same, $\text{Sd}[\mathcal{E}_i] = \sigma_e$.

Exercise 5.2. Open the file `flat_prices.csv`, that we have earlier worked on. We will start by studying the relation between flat *sizes* and selling *prices*. Remember the prices in the data does not reflect current price levels.

- Let us first do it simple, and ignore the fact that the prices are collected in three towns with potentially different housing market parameters. Start by producing a scatterplot of `area` vs `price`.
- Run a regression with prices as dependent variable, and square meter size as the independent variable. Find the estimated coefficients b_0, b_1 , and 95% confidence intervals for the true regression parameters β_0, β_1 .
- Reproduce the scatterplot from part a. and add the regression line estimated in part b.
- How would you interpret the value of the parameter β_1 (or the estimate b_1) in this regression model? (Hint: what would be the difference between a flat of size 93 and a flat of size 94 square meters according to the model?)
- What part of the regression results shows that the square meter size really does have an impact on the prices? You may be able to point out two related ways to see this from the output.
- What is the R^2 value for this regression? What does this number tell us about the impact of square meter size on selling prices?

- g. What is the value of the error term standard deviation S_e for this regression? Try to explain what this number represents in the regression.
- h. What would be your best guess for the price of a 100 m^2 flat? How can the S_e from the previous question be used to evaluate the precision of your guess?

Exercise 5.3. Continue working on the flat prices data. We will do a little preliminary analysis to see whether there are any detectable systematic differences between the house markets in the three towns involved in the data. We recall that the coding of the variable `town` is as follows.

- `town = 1` for Molde
- `town = 2` for Kristiansund
- `town = 3` for Ålesund

- a. Use the following code to make a scatterplot of prices vs square meter sizes, with different color dots for Molde, Kristiansund, Ålesund. The code assumes your dataframe is called `flp`, just change that to whatever you call it.

```
#make plot, col = "color", pch = "point character",
#we want solid dots (pch = 20)
with(flp, plot(area, price, col = town, pch = 20,
               main = "Price vs Area"))
legend("topleft", legend=c("M", "K", "A"), col = c(1, 2, 3), pch = 20)
```

Do you see anything indicating a difference in the three markets?

- b. We want to have a similar plot with one regression line for each town. This can be achieved with `base R` plotting, but is much better done with `ggplot`. Try the code below, you need to run `install.packages("ggplot2")` once to make this work.

```
library(ggplot2)
ggplot(flp, aes(x = area, y = price, color = as.factor(town))) +
  geom_point(size = 2) +
  geom_smooth(method = lm, se = FALSE)
```

Do we see any interesting differences in the lines?

- c. Use the `subset` option to run regression analysis on the price vs size relation, with separate estimates for each town. Make sure to produce 95% confidence intervals for coefficients. E.g. for Molde, you can do

```
#run regression for only Molde data, look at coefs and confidence intervals.
regMolde <- lm(price ~ area, data = subset(flp, town == 1))
coef(regMolde)
confint(regMolde)
```

- d. Find the estimated (marginal) square meter price (β_1) in each town.
- e. Just judging from the confidence intervals, do we find evidence for any real differences in the square meter prices between any pair of towns?

Exercise 5.4. One aspect of Supply Chain Management is decisions relating to pricing (i.e. to set prices optimal for your business) of products and services. Here is an example that

illustrates possible use of regression analysis as a tool to support the decision making. Suppose you are the manager of WaterWorld, a center with all kinds of water-sports and comfort facilities. For simplicity let's assume there is one single admission price called x . If you put x too high, no customers will come, if too low, many customers will come, but you will not earn any profit from them. The crucial relation is between the *price* on one hand, and the *demand* on the other. Demand in this case means number of tickets sold in a period. Assume the price x is kept fixed for one week at a time, and that we observe the demand y in the corresponding week. We assume here a linear relation in regression sense

$$y = \beta_0 + \beta_1 x + \mathcal{E}$$

In the data file 'WaterWorld.csv' you find data for demand (number of sold tickets) and prices for 35 weeks. Assume WaterWorld is in Norway, with prices in NOK. These data can be used to estimate a linear relation in line with the model.

- Make a scatterplot with regression line to visualize the relation. Comment on the use of a linear model. What would economic theory suggest for the relation in this case?
- Write $y = a - bx$ for the estimated regression line. Find values for a and b .
- How do you interpret the estimate b ?
- What can the estimated model tell us about a situation where we put the price to 0? Will we expect to "sell" a tickets at 0 NOK?
- Comment on the values of R^2 and S_e . What do they tell us about the relation?
- Find a prediction for the number of sold tickets and an approximate 95% margin of error if we put the price to $x = 110$.
- Let $P(x)$ denote the expected profit (Here we mean operational profit, i.e. before including interest, taxes and other financial cost/income into the accounting). This profit is simply the operational income minus the operational cost. Assume the following situation
 - The expected number of customers is as estimated by the regression, i.e. $y = a - bx$.
 - Each customer generates an expected income of $x + s$, where s is additional income from sale of food and drinks inside WaterWorld.
 - Each customer generates an expected cost of c , due mainly to energy cost for showers.
 - There is a fixed cost f covering salaries, energy, etc. that is independent of the number of customers.
 - These are the only operational income and cost factors.

Show that $P(x)$ can be expressed as

$$P(x) = -bx^2 + (a - bs + bc)x + as - ac - f$$

- Try to show that the profit $P(x)$ is maximized by

$$x = x^* = \frac{a - bs + bc}{2b} .$$

- Assuming that $s = 10, c = 5, f = 150000$, find the optimal price x^* , as well as the expected number of customers and the expected profit, given that price. What would be lost if using $x = 110$ instead of x^* ? You need no advanced math.
- The price x^* may not work well in practice because it is a decimal number. Suppose we want the price to be some whole multiple of 5 NOK. Which is more profitable: Rounding x^* *up or down* to the nearest multiple of 5?

Chapter 6

Linear multiple regression models

Exercise 6.1. The data file for this exercise is `Wages.csv`. You'll find it in the usual place. The file contains data for hourly wages and possibly related variables for $n = 1289$ US workers. The data is from 1995, worker ages range from 18 to 65. In this exercise we will only use a few of the involved variables, and study their possible effect on wage levels. Briefly stated, we restrict attention to the continuous variables

- $Y = \text{wage}$ = hourly wage in US\$.
- $X_1 = \text{education}$ = total number of years of schooling.
- $X_2 = \text{exper}$ = estimated years of work experience.
- $X_3 = \text{age}$. We should note that the X_2 variable was not directly observed as exact years of work experience, but rather computed by the formula

$$X_2 = X_3 - X_1 - 6 .$$

- Try to explain why the formula for X_2 makes (some) sense. For what workers will the formula give a correct value for the `exper` value? For what workers will it give a wrong value? Do you think there could be more correct values for females or males with this formula, or would it be the same for both genders?
- One might be tempted to estimate a linear regression model using Y as dependent variable, and X_1, X_2, X_3 as independent, so as to capture the simultaneous effect of *age*, *education* and *work experience* on wage levels. Why is this not possible with these data? Try with R and see what you get...
- Let's define model A as the simple model using only *education* as a wage driver, formally:

$$Y = \beta_0 + \beta_1 X_1 + \mathcal{E} .$$

Estimate parameters for this model, find the R^2 . Also find a 95% confidence interval for the β_1 parameter.

- In model B, we add the variable X_2 to the equation, to get

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \mathcal{E} .$$

Does it seem that both X variables are significantly affecting Y , are the signs of the effects as expected? What happens to the R^2 when adding X_2 ? Does the inclusion of X_2 appear to change the effect of X_1 substantially? Why or why not?

- e. What is the estimated value of an extra year of education/work experience according to model B? (In terms of US\$ per hour for the worker).
- f. If Maria and Jane has work experience of 10 and 15 years respectively, and both have 12 years of education, find forecasts for their wages, with 95% approximate error margins.
- g. Compared to Maria, will model B predict a different wage for Jim, who has the same education and experience? Why or why not? (You are allowed to assume Maria is female and Jim is male.)

Exercise 6.2. Find the file `used_cars.csv` in Canvas. Try to run R regressions with the linear models A, B from the section “omitted variable bias” in the compendium, with the variable definitions given there. Make sure to produce 95% confidence intervals for coefficients.

- a. Write down the two estimated models.
- b. Verify the radical differences in the estimates of the mileage’s effect on prices.
- c. In your own words, try to explain what the β_3 parameter means for the two models.
- d. In the two models, the confidence intervals for β_3 are far from overlapping. Does it mean any of the intervals are wrong?

Exercise 6.3. Using a linear model like those in exercise 6.2 means we assume prices of aging cars fall by the same *amount* each year and by each 1000Km driven. A more realistic description would be to assume prices falling by a certain *percentage* each year, and a certain percentage by each 1000Km driven. Try to formulate a model representing percentage changes, using the three X variables. You could use p, q as the average percentage drop in prices. Note: The resulting model will be nonlinear!

Chapter 7

Non-linear regression models.

Exercise 7.1. In this exercise, you should try to work along the lines of example outlined in the section on dummy variables in chapter 7. In the second part, you will go even further with the data material. The data are in `flat_prices.csv`.

- a. Find and read your copy of the data file. For each of the three towns, compute dummy variables. Use the names
 - DMOL for Molde
 - DKSU for Kristiansund
 - DASU for Ålesund

(So that $DMOL = 1$ for all flats in Molde, 0 otherwise etc.)

- b. Run the model that explains most price variation - without taking town differences into account. Include only significant x variables.
- c. As a variation to the example in the text, try to include town dependency, but use Kristiansund as the reference category (reference market). Find the estimated model that include all - and only - significant x variables. Call this model “PRK” (Prices with Reference Kristiansund).
- d. Call the model from the text (where Molde was the reference category “PRM”. Is there any differences in explanatory power/prediction quality between PRK and PRM?
- e. Compare the coefficient estimates for the variables that are common to PRM and PRK. Are there large differences? Why are the constants so different? Find the connection between the two constant estimates and the estimated coefficient for DMOL in model PRK.
- f. Consider an example flat A with the following characteristics
 - area = 90 sqm, #rooms = 3, standard = 3
 - town = 1 (Molde), distance to center = 3 km
 - location = 8, age = 25, rent = 2500 NOK.

Produce price forecasts for this flat using PRK and then using PRM. The price forecasts should be similar. Are they?

- g. What are the estimated average difference in prices between Molde and Kristiansund?

Between Ålesund and Kristiansund? And between Molde and Ålesund? Which are significant?

- h. (Theory question, use paper and pen.) Suppose Y = price and X_1 = area in the model PRK, and X_i denotes other significant original variables. Then the model reads

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_M DMOL + \beta_A DASU + \mathcal{E}.$$

Here town dependency is modeled only as a direct level shift. The marginal square meter price β_1 is assumed the same for all towns. Still using Kristiansund as reference market, write down a model allowing different square meter prices in different towns. (Hint: See the section on interaction effects.)

- i. If you found a satisfactory answer to the above question, try to use R to estimate parameters for the model. Is there any evidence of different marginal square meter prices between towns?
- j. The variable describing the number of rooms is an ordinal variable with relatively few values. Including this variable in a linear fashion shows no significant effect (when area is also included). Think about ways to include the number of rooms information in the model, using dummy variables. One could use rooms = 1 as reference category. Then use dummy variables e.g. as follows
- R2 indicating rooms = 2,
 - R3 indicating rooms = 3,
 - R4 indicating rooms = 4,
 - R5 indicating rooms ≥ 5 .

On paper sketch how you could implement general price corrections depending on number of rooms. Also think about how you could model - and investigate - possible different square meter prices depending on number of rooms. If you have time, try to do some experiments in this direction with R, using the available data.

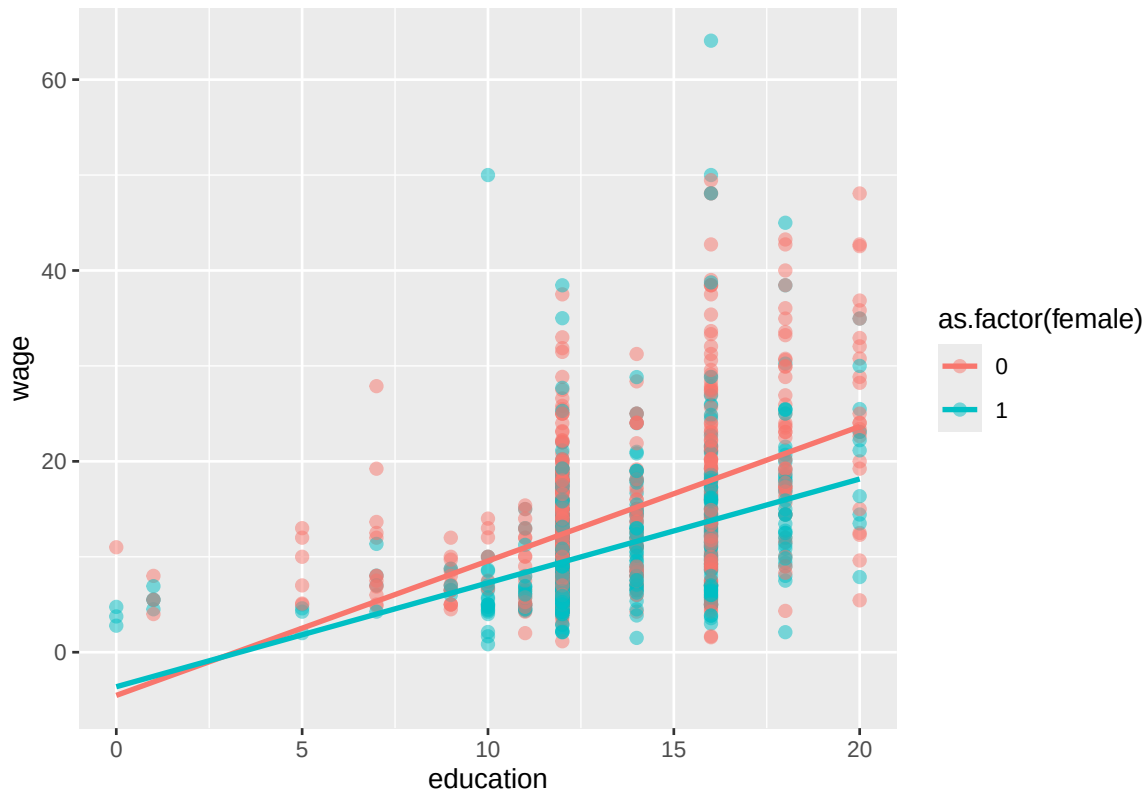
- k. As a purely theoretical exercise, think about how we could now for instance model a *separate marginal square meter price for 3-rooms flats in Molde*. This shows that using dummy variables in modeling interaction effects gives possibilities for estimating **very specific** effects in e.g. an economic market. This level of detail is not relevant in this example, but could be in other examples. Note also that there are only 16 3-rooms flats in Molde in the data set. When the model becomes more detailed, we risk that the estimated effects are backed by very small pieces of the data set. That is also a reason for not having individual dummy variables for rooms = 5, 6, 7 and 8, since there are very few observations here.

Exercise 7.2. We refer to the data in exercise 6.1, `wages.csv`. Read this file into a dataframe. The models we previously considered did not include information on gender. The data contains an indicator variable `female` which is 1 for all female workers, 0 otherwise.

- a. Along the lines of what we did for flat prices, try to include `female` along with *education* and *experience* in a regression model. What - according to this model - is the difference between male and female wage levels?
- b. Make a scatterplot of wage vs. education, with colors based on gender. Something like below will work using `ggplot`, assuming the dataframe is `wages`.

```
library(ggplot2)
ggplot(wages, aes(x = education, y = wage, color = as.factor(female))) +
  geom_point(size = 2, alpha = 0.5) +
  geom_smooth(method = lm, se = FALSE)

## `geom_smooth()` using formula = 'y ~ x'
```



What do we see from the regression lines?

From the plot it appears that males are gaining more from extra years of education than females. This would be an interaction effect. We can estimate interaction effects in some different ways, One way would be to compute a new variable that is the product of female and education and add this in the regression. As is often the case, R provides a direct way: So suppose we want to estimate separate effects of education and gender, *plus* the interaction of those two, in addition we want the experience in the model. We can use the following code.

```
wagereg <- lm(wage ~ education*female + exper, data = wages)
```

In the summary, the term `education:female` represents the interaction effect, i.e. the difference in the education effect between males and females.

- Try to run this regression. What is the estimated wage effect of one extra year of education for males? and for females?
- Compare forecast wages for Jim and Maria from exercise 6.1 with this model.

- e. The results from these models appear to show a wage discrimination of females, since with the same education and experience we find clear differences in the average wages. Could (parts of) these effects be “false” because of omitted variable bias? Think about some other factors deciding wages that are not included here, and that may be different for males/females. (Remember that the data are from USA 1995).
- f. Include the interaction between **exper** and **female**. Is this interaction effect significant?

Chapter 8

Logarithmic Transformations

Exercise 8.1. This is a theoretical *modeling* exercise, relating to demand modeling. Suppose we have daily recorded data for the variables

- y = number of airline passengers leaving from Molde airport with destination Oslo.
- x = average price of tickets used that day.
- w = day of week, 1 = Monday, ..., 7 = Sunday.

in all models, just write the relation you suggest between variables, ignore random terms.

a. Write the general constant elasticity model for the price - demand relation in this case.
b. Write a model that would allow statistical testing and estimation of day-to-day demand variations. It is assumed that demand is stable Tuesday to Thursday, peaking on Friday, Sunday and Monday, and low on Saturday. Your model should capture at least these effects. Write also the logarithmic transformed model. Make sure it is linear. * One could suspect that most travelers will try to avoid going on Saturday. Perhaps mainly very price-sensitive customers will go on Saturday if they find cheap tickets. Price sensitivity is to a large extent measured by the price elasticity. If you go back to your original model from a., it assumes price elasticity is the same on every weekday. Try to modify the model to allow estimating a change in price elasticity on Saturdays. Can this model be linearized by logarithmic transformation? Does it seem necessary to calculate new variables to estimate the possible change in elasticity?

Exercise 8.2. Find the file `meat_brands.sav` in Canvas. The data is described as follows. A supermarket sells two brands of ground meat, A and B. Brand A is cheaper, and brand B is considered a quality brand. Imagine the supermarket have fixed prices for each week, then possibly change it between weeks. One can imagine at least three types of buyers of the product:

- Highly price-conscious ones, who care mainly about the price of the products.
- “Opportunists”, who will buy the quality brand unless price difference is large.
- Quality seekers (or people with too much money) who always buy brand B (almost) regardless of prices.

Weekly sale volumes in kg, of each brand are recorded along with prices for each brand. Let y_A, y_B be the sale volumes for brand A, B, while x_A, x_B are prices in the same week. The

classical demand models state

$$y_A = c_A x_A^{\alpha_A} x_B^{\alpha_B}, \quad y_B = c_B x_A^{\beta_A} x_B^{\beta_B}.$$

- a. In the model for y_A , if any of the parameters α_A, α_B are different from 0, what sign would you expect on each? What would it mean if $\alpha_B = 0$ in the model?
- b. Theory tells us that the relation between logarithms of the variables is linear. Use linear regressions in R to estimate parameters α_A etc. for the two models. Take out eventual nonsignificant variables and rerun regressions if necessary.
- c. Discuss the differences you read from the estimated elasticities. In what way does the price of brand A (B) affect the sale of brand A? In what way does the price of brand A (B) affect the sale of brand B? Could some of the effects be explained by the possibly different customer groups as mentioned above?
- d. Difficult: Discuss whether (and how) advertisement and short-term-low-price campaigns e.g. for meat A could invalidate the models we have used here. This means, do you see possible market effects we can not estimate with our simple models? Does it matter that almost all households in the town has a deep-freezer? Could some modifications of the models help?